

AMAT113 Calculus II

Homework 3

Area Between Curves and Determining Volumes

Submission Instructions

- Homework is due **Sunday at 11:59 PM Eastern Time.**
- Please print this PDF and complete the problems neatly by hand whenever possible.
- If printing is not available, completing the work on loose-leaf paper is acceptable.
- Show all necessary work. Answers without supporting work may receive reduced credit.
- Scan all completed pages, combine them into one PDF file, and submit the PDF through **Brightspace before the deadline.**
- Before submitting, confirm that every page is included, upright, legible, and in the correct order.

Part I: Area Between Curves

1. For each of the following problems, graph the equations and shade the area of the region between the curves. Determine its area by integrating over the x -axis.

(a) $y = x^2$ and $y = -x^2 + 18x$

(b) $y = e$, $y = e^x$, and $y = e^{-x}$.

Part I: Area Between Curves (continued)

2. For each of the following problems, graph the equations and shade the area of the region between the curves. If necessary, break the region into sub-regions to determine its entire area.

(a) $y = 12 - x$, $y = \sqrt{x}$, and $y = 1$

(b) $y = x^3$ and $y = x^2 - 2x$ over $x \in [-1, 1]$.

Part I: Area Between Curves (continued)

3. For each of the following problems, graph the equations and shade the area of the region between the curves. Determine its area by integrating over the y -axis.

(a) $y = x^3$ and $x = 3y - 2$

(b) $x = |y|$ and $2x = -y^2 + 2$.

1. Draw an outline of the solid and find the volume using the slicing method.
 - (a) The base is the region under the parabola $y = 1 - x^2$ in the first quadrant. Slices perpendicular to the xy -plane and parallel to the y -axis are squares.
 - (b) The base is the area between $y = x$ and $y = x^2$. Slices perpendicular to the x -axis are semicircles.

Part II: Determining Volumes (continued)

2. Draw the region bounded by the curves $y = e^x + 1$, $x = 0$, $x = 1$, and $y = 0$. Use the disk method to find the volume when the region is rotated around the x -axis.

3. Draw the region bounded by the curves $y = \sqrt{4 - x^2}$, $x = 0$, and $y = 0$. Find the volume when the region is rotated around the y -axis.

Part II: Determining Volumes (continued)

4. Draw the region bounded by the curves $y = \sqrt{x}$, $x = 4$, and $y = 0$. Use the washer method to find the volume when the region is rotated around the y -axis.

5. Draw the region bounded by the curves $x = \sqrt{9 - y^2}$, $x = e^{-y}$, $y = 0$ and $y = 3$. Use the washer method to find the volume when the region is rotated around the y -axis.